

MDHGNN project description

Copper futures price forecasting based on yield modeling

I. Project positioning

This project implements multi-scale dynamic hypergraph neural network MDHGNN for copper futures weekly frequency prediction. Direct prediction target of the current code is the future yield, not directly price, is not a separate classification move up/down. The price forecast result is obtained by restoring the last period price and the forecast yield; The direction accuracy is derived from the sign of the predicted return rate.

Caliber	Actual meaning in the project
Anticipate the goal directly	Vector of future h-weekly returns.
Price forecasts	Use the last price P_{t-1} and the predicted rate of return to restore the future price, and then calculate the price error.
Direction forecasting	Not training a classifier separately; The direction is obtained by predicting whether the return is greater than 0.
Paper/report recommended title	Copper futures Price Forecasting based on Yield modeling.

Ii. Data description

Raw data in the data/raw /, copper futures for the target assets, WTI crude oil, gold, silver, and the dollar index as the external market variables. Current delivery results are in a weekly frequency setting.

Project	Instructions
Copper futures raw data	Daily frequency, 1996-03-11 to 2016-01-29, 5000 lines in total.
Circular frequency after resampling	1996-03-15 to 2016-01-29, 1038 lines.
Join indicators and delete after missing	1,013 lines from 1996-09-06 to 2016-01-29.

Number of features	30, including OHLCV, Change, technical indicators, and external market variables.
The window length	Cycle frequency experiment W=12.
Main prediction step size	h=1 and h=2.

Data partitioning

Collection	Row index	Date range	Line count	Actual target sample
Training set	0-708	1996-09-06 to 2010-04-02	709	697
Validation set	709-809	2010-04-09 to 2012-03-09	101	101
Test set	810-1012	2012-03-16 to 2016-01-29	203	202

To prevent data leakage

1. The training, validation, test segmentation in chronological order, do not do random divided across a period.

Z-score normalization uses only the mean and standard deviation of the training set and applies it to the validation and test sets.

3. The input window for the TTH sample contains only $x_{\{t-W\}}, \dots, x_{\{t-1\}}$, does not contain the target price.

External variables are aligned by date and resampled to the cycle frequency; In this examination, the number of missing external variables was 0 after merging.

3. Prediction objective and formula

$$\begin{aligned}
X_t &= [x_{\{t-W\}}, \dots, x_{\{t-1\}}] \\
r_{\{t,j\}} &= (P_{\{t+j-1\}} - P_{\{t-1\}}) / P_{\{t-1\}}, \quad j = 1, \dots, h \\
f_{\theta}(X_t) &= \hat{r}_t = [\hat{r}_{\{t,1\}}, \dots, \hat{r}_{\{t,h\}}] \\
\hat{P}_{\{t+j-1\}} &= P_{\{t-1\}} * (1 + \hat{r}_{\{t,j\}}) \\
\hat{d}_{\{t,j\}} &= 1(\hat{r}_{\{t,j\}} > 0)
\end{aligned}$$

Fourth, model structure

MDHGNN contains two paths: sequential path using linear projection, location of the code, pooling GRU helped and attention; Structure features as a hypergraph path to 30 input nodes, through the characteristics of hypergraph convolution learning

higher-order relationship between variables. After the two paths are fused by learnable weights, the yield rate is output by the prediction head.

The module	Files	instructions
Master model	models/mdhgnn.py	It includes feature node coding, feature hypergraph construction, structure pooling, time series path and fusion prediction.
Hypergraph convolution	models/hgcn_conv.py	To super, super while to realize node information dissemination and attention gating.
Temporal aggregation	models/temporal_aggregator.py	GRU plus attention pooling.
Predict the head	models/prediction_head.py	Two-layer MLP that outputs h-step yields.

Current three classes of hyperedges

The current active model only keep the following characteristics of three kinds of super edge, hypergraph structure around the correlation between input characteristics, field a priori and adaptive relationship building.

Types of Hyperedges	Construction method	Quantity
Correlation between soft edge	Linear projection for input window, calculation similarity between features, associated with softmax get of soft. Not Top - k, not threshold screen edge.	90
Domain prior hyperedges	Build fixed correlations by economic implications such as price, trend, momentum, volatility, volume, external market, and global group.	9
Adaptive hyperedges	Generated by the embedded feature nodes through can learn projection.	8
Aggregate	Nodes are 30 input features.	107

V. Baseline methods

Methods	Documentation	Instructions
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Naive	models/baselines/naive_baseline.py	The parameterless persistence baseline, where the predicted rate of return is always 0, assumes that the next price is equal to the last price. Code implementation for nn. The interface Module just for unified assessment.
GRU	models/baselines/gru_baseline.py	Two layers of GRU helped regression model.
MLP	models/baselines/mlp_baseline.py	Expand the input window and regress with a multi-layer fully connected network.
Transformer	models/baselines/transformer_baseline.py	Transformer Encoder baselines.
LSTM/TCN	models/baselines/	Code reserved, comparison can be added as needed.

$$\hat{r}_{t,j}^{\text{Naive}} = 0$$

$$\hat{P}_{t+j-1}^{\text{Naive}} = P_{t-1}$$

Vi. Evaluation indicators

Categories	Metrics	Instructions
Price level	RMSE, MAE, MAPE, R2	Calculate using the reduced price P and $\hat{P}$.
Yield level	Ret_RMSE, Ret_R2	Ret_RMSE is expressed in base points.
Direction level	DA, DA_up, DA_down	Calculated by the sign of the predicted and true returns.
Strategy level	Sharpe, AnnReturn, MaxDrawdown	According to the sign (pred_return) form long-short direction after calculation.
Significance test	DM test	MDHGNN as reference, $DM_{stat} < 0$ means MDHGNN square error is lower.

Vii. Main experimental results

The following results were reproduced by the current code on 2026-06-11, weekly frequency, seeds 42, 123, 456. results files are located at results/tables/ and plots at results/figures/.

h	Model	RMSE		R2		Ret_RMSE		DA	
1	MDHGNN	0.0726	±	0.9748	±	237.36	±	53.1	±
		0.0006		0.0004		2.01		5.4	
1	Naive	0.0722	±	0.9751	±	236.06	±	56.9	±
		0.0000		0.0000		0.00		0.0	
2	MDHGNN	0.1133	±	0.9379	±	314.39 + / -		44.4	±
		0.0051		0.0056		15.21		2.1	
2	Naive	0.1063	±	0.9454	±	296.48	±	57.5	±
		0.0000		0.0000		0.00		0.0	

Viii. Ablation experiments

In the ablation experiment, the mask ablation caliber was used: each seed trained the complete MDHGNN once, and then shielded a hyperedge channel or timing path under the same weight in the inference stage. The result has been written results/tables/ablation_hl_manifest json.

Variants	RMSE		R2	
Full MDHGNN	0.0726	±	0.9748	±
	0.0006		0.0004	
w/o Corr-HE	0.1256	±	0.8975	±
	0.0751		0.1094	
w/o Domain-HE	0.1251	±	0.8984	±
	0.0746		0.1083	
w/o Adapt-HE	0.1251	±	0.8984	±
	0.0746		0.1082	
w/o GRU	0.7973	±	-2.7114	±
	0.3740		2.7119	

Ix. Mode of operation

Installation Dependencies:

```
pip install -r requirements.txt
```

Single model training:

```
python main.py --mode train --frequency weekly --horizon 1 --seed 42
```

The main contrast experiment:

```
python experiments/run_all.py --frequency weekly --horizons 1 2 --seeds 42 123 456
```

Official ablation experiments (do inference masking after Full training) :

```
python experiments/run_masked_ablation.py --frequency weekly --horizon 1 --seeds 42 123 456
```

X. Project structure

Files/directories	instructions
config.py	Centralized management model and data, and the training parameters.
main.py	Single model training entry.
data/	Data reading, resampling, feature engineering, preprocessing, and Dataset.
models/	MDHGNN, hypergraph convolution, prediction head, and baseline method implementation.
experiments/run_all.py	For the main comparison experiment, results_h1.json, results_h2.json, DM test and graph were generated.
experiments/run_masked_ablation.py	Ablation experiments were officially masked to generate per-seed, multi-seed, manifest, and ablation maps.
results/tables/	JSON metrics, DM test, ablation summary, and experimental manifest.
results/figures/	PNG, PDF, SVG charts.
tests/	Unit tests, covering chart output, DM JSON, and ablation variant forward propagation.